Paper introduction

Ultra-fast Screen-Space Refractions and Caustics via Newton's Method

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* All figures without annotation in this slide are quoted from the same paper

Overview

- Realtime screen-space refractions and caustics [1]
  - **Avoid ray marching by using Newton's method** for intersection tests
  - Achieve convergence **only in 2-6 iterations**, enabling real-time performance (ray marching typically requires 100 steps)

Figure 1: teaser image: A boat scene

Background and Context

- In computer graphics, **refractions and caustics are important** for transparent materials
  - like water and glass
- Caustics rendering using full ray tracing (path tracing) is **completely impractical for real-time applications**…
  - Even path tracing, it is too difficult to converge in low spp
- Traditional methods rely on **screen-space ray marching**.
  - computationally expensive → not suitable for real-time applications

complement

Key Idea: Newton's Method

- **Newton's method** is an iterative root-finding algorithm that can be used to find the intersection of rays with surfaces
  - It converges quadratically near the root, which allows for fast convergence in a few iterations

$$s_{\{n+1\}} = s_n - \frac{f(s_n)}{f'(s_n)}$$

Newton's method iteration formula

- The authors adapt Newton's method to solve the ray-surface intersection problem in screen space

References

[1] C. Mayer, U. Assarsson, and E. Sintorn, “Ultra-fast Screen-Space Refractions and Caustics via Newton’s Method,” Journal of Computer Graphics Techniques (JCGT), vol. 15, no. 1, pp. 44–59, Apr. 2026, [Online]. Available: http://jcgt.org/published/0015/01/03/